Zheyu Chen

Graduate Researcher in Aerial Robotics and Control
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PROFILES

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RESEARCH INTERESTS

Autonomous flight and robotics **UAV flight control** **Over-actuated aerial vehicles**

SUMMARY

Graduate researcher in aerial robotics and control systems with hands-on experience in the design, modeling, and control of over-actuated UAV platforms. My research focuses on tricopter systems and advanced control methods for high-maneuverability aerial vehicles. I have developed and deployed multiple UAV platforms with real-world flight validation . I am particularly interested in UAV flight control, over-actuated aerial vehicles and autonomous flight.

EDUCATION

Sun Yat-sen University — B.Eng. in Aerospace Engineering • GPA 3.6/5.0
Aerospace Engineering — Shenzhen, Guangdong, China • 2020-2024

Sun Yat-sen University — M.S. in Aerospace Engineering • GPA 4.2/5.0
UAV control — Shenzhen, Guangdong, China • 2024-Present

PROJECTS

Over-Actuated Tricopter Platform — 2024-2025
A novel over-actuated tricopter featuring a servo-driven twisting and tilting mechanism is presented.

A Passively Articulated Overactuated Tricopter — 2024-2025
A passively articulated over-actuated tricopter is presented along with a Two-Step INDI (TS-INDI) control framework.

SKILLS

MATLAB
Advanced

Simulink
Advanced

C++
Advanced

PX4
Advanced

Solidworks
Advanced

PUBLICATIONS

Design and Flight Control of a Novel Thrust-Vectored Tricopter Using Twisting and Tilting Rotors — 2025
2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)